

Smart farming

While precision farming was mainly based on the use of GPS sensors for auto-steering of tractors and harvesters, the focus of smart farming has now shifted towards smart treatments like deciding the amount of fertiliser that should be applied, or using plant protection resources for optimal crop development in a particular area in the field.

Data- and information-driven agriculture is a general trend in smart farming. Earth observation (EO) and navigation satellites use data collected from ground sensors to help farmers decide how, when and where to allocate resources for best results.

Satellites are mounted with multiple sensors. Information on current growth status and development of crops at each location in the field is obtained from the satellite(s) through various plant conditions and parameters such as biomass and chlorophyll content during growing season. Such data help farmers update crop status for specific plant protection and fertilisation measures. Farmers can also monitor various parameters on smartphones or tablets on-the-go.

Cognitive sensing technology. Precision farming can be enhanced using cognitive sensing technology. For example, IBM Watson can adapt to consider new parameters and uncover correlations between various parameters, using machine learning.

A cognitive system with advanced sensor technology provides real-time feedback like temperature, humidity, light, soil pH, precipitation, irrigation timing and so on. It can help identify actual problems affecting the crops. Cloud-based services with a cognitive system learn from results all around the world in real time, so that each season brings better insights, solutions and recommendations for better yields.

Smart spraying technology. To combat weeds and undesired plants that affect crops, herbicides are applied over large surface areas, covering crops and fields. Bosch and Bayer are developing smart spraying technology that uses camera sensors to distinguish between weeds and crops. This way only weeds are targeted, and crops and the environment are protected.



Fig. 3: Smart farming (Credit: <https://link.springer.com>)



Fig. 4: Virtual reality system for chickens (Credit: www.secondlivestock.com)

Plant-specific herbicide application requires sensor systems for plant recognition and differentiation. However, it is not easy to recognise small weeds in the early stages of development. With modern technology, programmable true-colour sensors can be used for real-time recognition and identification of individual weeds and crops. This can be achieved by collecting data based on the reflection properties of plants, and natural and artificial backgrounds. Crops and weeds are recognised using mathematical algorithms and decision models based on such data.

The IoT for agriculture

The IoT for agriculture, or IoT4Ag, is a new area of technology that holds tremendous potential for improving global food production. IoT sensors can be deployed on the ground or in water to collect data on target inputs, such as soil moisture and crop health. Collected data are stored on a server or cloud system wirelessly, and can be easily accessed by farmers via the Internet on tablets and cellphones.

IoT applications include farm vehi-

cle tracking and livestock monitoring, and in arable farming, agri-food supply chain, green house monitoring and other farm operations. Farmers can remotely control activities in the farm through connectivity options like Nano Ganesh, which is a mobile-

based remote controller for water pumps.

There are IoT devices from Smart Elements, wherein a variety of sensors are deployed in a farm that report back to an online dashboard. Farmers need not visit the farm, thus saving time and money by making fast and informed decisions based on real-time conditions.

Adoption of IoT solutions for agriculture is constantly growing. Cellular companies are seeking business opportunities in farm IoT applications. They are developing integrated systems that combine sensors and communication systems. IoT sensors and cloud computing will drastically improve the quality of data flow, which will, in turn, help farmers make better decisions.

Digital livestock farming

As population increases, demand for milk and meat also increases. Digital and sensor technologies are being deployed in livestock management. Precision livestock farming uses advanced technologies, such as microfluidics, sound analysers, image-detection techniques, sweat and saliva sensing, serodiagnosis and others, to optimise the contribution of each animal.

Smarter production, proper nutrition as well as increased animal welfare have the potential to decrease the impact of livestock farming on resources. Many companies are working on the welfare of livestock animals, while others are working on animal health management.

With such technology, livestock such as chickens are treated as humanely as possible. Computer interfaces and virtual environments